ΔΙΚΤΥΑ ΥΠΟΛΟΓΙΣΤΩΝ (5η Εργαστηριακή Άσκηση)

ΕΠΩΝΗΜΟ:ΚΙΟΥΡΤ ΜΕΧΜΕΤΑΛΗ

ΟΝΟΜΑ:ΑΧΜΕΤ

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Ερωτήσεις (1ο Κομμάτι)

1. Βλέπουμε ότι υπάρχει η επικοινωνία μεταξύ των switches.

Interface R1

```
[admin@R1] > ip/address/print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE
#   ADDRESS             NETWORK        INTERFACE
0   10.255.255.1/32     10.255.255.1   loopback0
1   108.46.72.1/30      108.46.72.0    ether2
2   108.46.72.5/30      108.46.72.4    ether3
3 D 192.168.171.144/24  192.168.171.0  ether1
[admin@R1] > ping 108.46.72.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.2                                56  64 1ms987us
    1 108.46.72.2                                56  64 558us
    2 108.46.72.2                                56  64 580us
    3 108.46.72.2                                56  64 536us
    4 108.46.72.2                                56  64 839us
    sent=5 received=5 packet-loss=0% min-rtt=536us avg-rtt=900us max-rtt=1ms987us

[admin@R1] > ping 108.46.72.6
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.6                                56  64 544us
    1 108.46.72.6                                56  64 392us
    2 108.46.72.6                                56  64 449us
    3 108.46.72.6                                56  64 478us
    sent=4 received=4 packet-loss=0% min-rtt=392us avg-rtt=465us max-rtt=544us

[admin@R1] >
```

Interface R2

```
[admin@R2] > ip/address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS          NETWORK       INTERFACE
0 10.255.255.2/32  10.255.255.2  loopback0
1 108.46.72.2/30   108.46.72.0   ether1
2 108.46.72.9/30   108.46.72.8   ether2
3 108.46.72.13/30  108.46.72.12  ether3
[admin@R2] > ping 108.46.72.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.1                                56  64 401us
    1 108.46.72.1                                56  64 551us
    2 108.46.72.1                                56  64 558us
    3 108.46.72.1                                56  64 549us
    4 108.46.72.1                                56  64 495us
    sent=5 received=5 packet-loss=0% min-rtt=401us avg-rtt=510us max-rtt=558us

[admin@R2] > ping 108.46.72.10
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.10                               56  64 382us
    1 108.46.72.10                               56  64 629us
    2 108.46.72.10                               56  64 543us
    3 108.46.72.10                               56  64 508us
    4 108.46.72.10                               56  64 583us
    sent=5 received=5 packet-loss=0% min-rtt=382us avg-rtt=529us max-rtt=629us

[admin@R2] >
```

Interface R3

```
[admin@R3] > ip/address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
0 10.255.255.3/32   10.255.255.3  loopback0
1 108.46.72.6/30    108.46.72.4   ether1
2 108.46.72.10/30   108.46.72.8   ether2
[admin@R3] > ping 108.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.5                                56  64 1ms269us
    1 108.46.72.5                                56  64 456us
    2 108.46.72.5                                56  64 609us
    3 108.46.72.5                                56  64 497us
    sent=4 received=4 packet-loss=0% min-rtt=456us avg-rtt=707us max-rtt=1ms269us

[admin@R3] > ping 108.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.9                                56  64 403us
    1 108.46.72.9                                56  64 497us
    2 108.46.72.9                                56  64 501us
    3 108.46.72.9                                56  64 959us
    4 108.46.72.9                                56  64 557us
    sent=5 received=5 packet-loss=0% min-rtt=403us avg-rtt=583us max-rtt=959us

[admin@R3] >
```

2. Υπάρχει, που σημαίνει ότι έχουμε διαδρομή στο εξωτερικό κόσμο.

```
[admin@R3] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY              DISTANCE
DAo  0.0.0.0/0         108.46.72.5%ether1        110
DAc  10.255.255.3/32   loopback0                   0
DAo+ 108.46.72.0/30    108.46.72.9%ether2        110
DAo+ 108.46.72.0/30    108.46.72.5%ether1        110
DAc  108.46.72.4/30    ether1                      0
DAc  108.46.72.8/30    ether2                      0
[admin@R3] >
```

```
[admin@R1] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, o, d, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY             DISTANCE
DAd  0.0.0.0/0         192.168.171.2              1
DAc  10.255.255.1/32   loopback0                  0
DAc  108.46.72.0/30    ether2                     0
DAc  108.46.72.4/30    ether3                     0
DAo+ 108.46.72.8/30    108.46.72.6%ether3       110
DAo+ 108.46.72.8/30    108.46.72.2%ether2       110
DAc  192.168.171.0/24  ether1                     0
[admin@R1] >
```

```
[admin@R2] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY              DISTANCE
DAo  0.0.0.0/0         108.46.72.1%ether1        110
DAc  10.255.255.2/32   loopback0                   0
DAc  108.46.72.0/30    ether1                      0
DAo+ 108.46.72.4/30    108.46.72.10%ether2       110
DAo+ 108.46.72.4/30    108.46.72.1%ether1        110
DAc  108.46.72.8/30    ether2                      0
DAc  108.46.72.12/30   ether3                      0
[admin@R2] >
```

3. Η επικοινωνία με το Debian για το **traceroute 108.46.72.1** και **traceroute 108.46.72.5** γίνεται κανονικά από το **DHCP-Server** με **IP 108.46.72.13**.

```
debian@debian:~$ traceroute 108.46.72.1
traceroute to 108.46.72.1 (108.46.72.1), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  5.723 ms  5.618 ms  5.606 ms

 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * *^C
debian@debian:~$ traceroute 108.46.72.5
traceroute to 108.46.72.5 (108.46.72.5), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  2.060 ms  2.024 ms  2.019 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * *^C
debian@debian:~$
```

4. Βλέπουμε ότι μόνο ο R1 μπορεί να κάνει το **ping 8.8.8.8** διότι έχει πρόσβαση στο εξωτερικό κόσμο.

```
[admin@R1] > ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 128 35ms146us
    1 8.8.8.8                                    56 128 32ms769us
    2 8.8.8.8                                    56 128 32ms910us
    3 8.8.8.8                                    56 128 35ms626us
    4 8.8.8.8                                    56 128 32ms338us
    5 8.8.8.8                                    56 128 34ms8us
    sent=6 received=6 packet-loss=0% min-rtt=32ms338us avg-rtt=33ms799us max-rtt=35ms626us

[admin@R1] >
```

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```
[admin@R2] > ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0                                                              no route to host
    1                                                              no route to host
    2                                                              no route to host
    3                                                              no route to host
    4                                                              no route to host
    sent=5 received=0 packet-loss=100%

[admin@R2] >
```

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```
[admin@R3] > ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0                                                              no route to host
    1                                                              no route to host
    2                                                              no route to host
    3                                                              no route to host
    4                                                              no route to host
    5                                                              no route to host
    sent=6 received=0 packet-loss=100%

[admin@R3] >
```

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5. Προσθέτοντας την στατική διαδρομή παρατηρούμε ότι στην R2 έχουμε μία νέα διαδρομή που θα μας αφήσει την πρόσβαση να υπάρχει η επικοινωνία του **ping 8.8.8.8** στο interface R3. Που η επικοινωνία του R3 δρομολογείτε μέσο του **108.46.72.4** και **108.46.72.8.**

```
[admin@R3] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY              DISTANCE
DAo  0.0.0.0/0         108.46.72.5%ether1        110
DAc  10.255.255.3/32   loopback0                   0
DAo+ 108.46.72.0/30    108.46.72.9%ether2        110
DAo+ 108.46.72.0/30    108.46.72.5%ether1        110
DAc  108.46.72.4/30    ether1                      0
DAc  108.46.72.8/30    ether2                      0
[admin@R3] >
```

Έτσι ώστε να υπάρχει η επικοινωνία στο ping 8.8.8.8

```
[admin@R3] > ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 127 33ms101us
    1 8.8.8.8                                    56 127 32ms416us
    2 8.8.8.8                                    56 127 33ms974us
    3 8.8.8.8                                    56 127 31ms338us
    sent=4 received=4 packet-loss=0% min-rtt=31ms338us avg-rtt=32ms707us max-rtt=33ms974us

[admin@R3] >
```

Ενώ όταν αφαιρέσουμε την διαδρομή R2-R3 βλέπουμε ότι ναι μεν υπάρχει η επικοινωνία αλλά μόνο μέσο του **108.46.72.4**

```
[admin@R3] > ip/route/print
Flags: D - DYNAMIC; I, A - ACTIVE; c, y - COPY; H - HW-OFFLOADED
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY    DISTANCE
DAc  10.255.255.3/32   loopback0         0
DAc  108.46.72.4/30    ether1            0
DIcH 108.46.72.8/30    ether2            0
[admin@R3] > ping 8.8.8.8
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 127 34ms871us
    1 8.8.8.8                                    56 127 33ms983us
    2 8.8.8.8                                    56 127 34ms218us
    3 8.8.8.8                                    56 127 33ms560us
    sent=4 received=4 packet-loss=0% min-rtt=33ms560us avg-rtt=34ms158us max-rtt=34ms871us

[admin@R3] >
```

Ερωτήσεις (2^{Ο} Κομμάτι)

1^{η} Ερώτηση

```
[admin@R1] > ping 108.46.72.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.1                                56  64 538us
    1 108.46.72.1                                56  64 48us
    sent=2 received=2 packet-loss=0% min-rtt=48us avg-rtt=293us max-rtt=538us

[admin@R1] > ping 108.46.72.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.2                                56  64 749us
    1 108.46.72.2                                56  64 623us
    sent=2 received=2 packet-loss=0% min-rtt=623us avg-rtt=686us max-rtt=749us

[admin@R1] > ping 108.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.5                                56  64 111us
    1 108.46.72.5                                56  64 51us
    sent=2 received=2 packet-loss=0% min-rtt=51us avg-rtt=81us max-rtt=111us

[admin@R1] > ping 108.46.72.6
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.6                                56  64 620us
    1 108.46.72.6                                56  64 645us
    sent=2 received=2 packet-loss=0% min-rtt=620us avg-rtt=632us max-rtt=645us

[admin@R1] > ping 108.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.9                                56  64 2ms856us
    1 108.46.72.9                                56  64 580us
    sent=2 received=2 packet-loss=0% min-rtt=580us avg-rtt=1ms718us max-rtt=2ms856us

[admin@R1] > ping 108.46.72.10
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.10                               56  63 2ms80us
    1 108.46.72.10                               56  63 1ms224us
    sent=2 received=2 packet-loss=0% min-rtt=1ms224us avg-rtt=1ms652us max-rtt=2ms80us

[admin@R1] > ping 192.168.5.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.5.1                                56  64 120us
    1 192.168.5.1                                56  64 46us
    sent=2 received=2 packet-loss=0% min-rtt=46us avg-rtt=83us max-rtt=120us

[admin@R1] > ping 192.168.5.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.5.2                                56  64 923us
    1 192.168.5.2                                56  64 530us
    sent=2 received=2 packet-loss=0% min-rtt=530us avg-rtt=726us max-rtt=923us
```

```
[admin@R2] > ping 108.46.72.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.2                                56  64 184us
    1 108.46.72.2                                56  64 115us
    sent=2 received=2 packet-loss=0% min-rtt=115us avg-rtt=149us max-rtt=184us

[admin@R2] > ping 108.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.5                                56  64 922us
    1 108.46.72.5                                56  64 806us
    sent=2 received=2 packet-loss=0% min-rtt=806us avg-rtt=864us max-rtt=922us

[admin@R2] > ping 108.46.72.6
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.6                                56  64 729us
    1 108.46.72.6                                56  64 551us
    sent=2 received=2 packet-loss=0% min-rtt=551us avg-rtt=640us max-rtt=729us

[admin@R2] > ping 108.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.9                                56  64 1ms280us
    1 108.46.72.9                                56  64 45us
    sent=2 received=2 packet-loss=0% min-rtt=45us avg-rtt=662us max-rtt=1ms280us

[admin@R2] > ping 108.46.72.10
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.10                               56  64 659us
    1 108.46.72.10                               56  64 513us
    sent=2 received=2 packet-loss=0% min-rtt=513us avg-rtt=586us max-rtt=659us

[admin@R2] > ping 192.168.5.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.5.1                                56  64 683us
    1 192.168.5.1                                56  64 521us
    sent=2 received=2 packet-loss=0% min-rtt=521us avg-rtt=602us max-rtt=683us
```

```
[admin@R2] > ping 108.46.72.13
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.13                               56  64 878us
    1 108.46.72.13                               56  64 38us
    2 108.46.72.13                               56  64 42us
    sent=3 received=3 packet-loss=0% min-rtt=38us avg-rtt=319us max-rtt=878us
```

```
[admin@R3] > ping 108.46.72.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.1                                56  64 1ms825us
    1 108.46.72.1                                56  64 517us
    sent=2 received=2 packet-loss=0% min-rtt=517us avg-rtt=1ms171us max-rtt=1ms825us

[admin@R3] > ping 108.46.72.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.2                                56  64 693us
    1 108.46.72.2                                56  64 644us
    sent=2 received=2 packet-loss=0% min-rtt=644us avg-rtt=668us max-rtt=693us

[admin@R3] > ping 108.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.5                                56  64 493us
    1 108.46.72.5                                56  64 589us
    sent=2 received=2 packet-loss=0% min-rtt=493us avg-rtt=541us max-rtt=589us

[admin@R3] > ping 108.46.72.6
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.6                                56  64 71us
    1 108.46.72.6                                56  64 42us
    sent=2 received=2 packet-loss=0% min-rtt=42us avg-rtt=56us max-rtt=71us

[admin@R3] > ping 108.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.9                                56  64 476us
    1 108.46.72.9                                56  64 547us
    sent=2 received=2 packet-loss=0% min-rtt=476us avg-rtt=511us max-rtt=547us

[admin@R3] > ping 108.46.72.10
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.10                               56  64 1ms66us
    1 108.46.72.10                               56  64 47us
    sent=2 received=2 packet-loss=0% min-rtt=47us avg-rtt=556us max-rtt=1ms66us

[admin@R3] > ping 192.168.5.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.5.1                                56  64 607us
    1 192.168.5.1                                56  64 714us
    sent=2 received=2 packet-loss=0% min-rtt=607us avg-rtt=660us max-rtt=714us

[admin@R3] > ping 192.168.5.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.5.2                                56  62 1ms500us
    1 192.168.5.2                                56  62 1ms641us
    sent=2 received=2 packet-loss=0% min-rtt=1ms500us avg-rtt=1ms570us max-rtt=1ms641us
```

```
[admin@R4] > ping 108.46.72.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.1                                56  64 469us
    1 108.46.72.1                                56  64 1ms116us
    2 108.46.72.1                                56  64 469us
    3 108.46.72.1                                56  64 436us
    4 108.46.72.1                                56  64 431us
    sent=5 received=5 packet-loss=0% min-rtt=431us avg-rtt=584us max-rtt=1ms116us

[admin@R4] > ping 108.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.5                                56  64 399us
    1 108.46.72.5                                56  64 424us
    2 108.46.72.5                                56  64 530us
    3 108.46.72.5                                56  64 494us
    4 108.46.72.5                                56  64 332us
    sent=5 received=5 packet-loss=0% min-rtt=332us avg-rtt=435us max-rtt=530us

[admin@R4] > ping 108.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 108.46.72.9                                56  63 3ms681us
    1 108.46.72.9                                56  63 1ms75us
    2 108.46.72.9                                56  63 1ms240us
    3 108.46.72.9                                56  63 1ms134us
    4 108.46.72.9                                56  63 1ms76us
    sent=5 received=5 packet-loss=0% min-rtt=1ms75us avg-rtt=1ms641us max-rtt=3ms681us

[admin@R4] > ping 109.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.5                                56  64 912us
    1 109.46.72.5                                56  64 33us
    2 109.46.72.5                                56  64 35us
    3 109.46.72.5                                56  64 30us
    sent=4 received=4 packet-loss=0% min-rtt=30us avg-rtt=252us max-rtt=912us

[admin@R4] > ping 109.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.9                                56  64 718us
    1 109.46.72.9                                56  64 528us
    2 109.46.72.9                                56  64 582us
    3 109.46.72.9                                56  64 471us
    4 109.46.72.9                                56  64 555us
    sent=5 received=5 packet-loss=0% min-rtt=471us avg-rtt=570us max-rtt=718us

[admin@R4] >
```

```
[admin@R5] > ping 109.46.72.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.1                                56  64 1ms796us
    1 109.46.72.1                                56  64 597us
    sent=2 received=2 packet-loss=0% min-rtt=597us avg-rtt=1ms196us max-rtt=1ms796us

[admin@R5] > ping 109.46.72.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.2                                56  64 95us
    1 109.46.72.2                                56  64 41us
    sent=2 received=2 packet-loss=0% min-rtt=41us avg-rtt=68us max-rtt=95us

[admin@R5] > ping 109.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.5                                56  64 536us
    1 109.46.72.5                                56  64 617us
    sent=2 received=2 packet-loss=0% min-rtt=536us avg-rtt=576us max-rtt=617us

[admin@R5] > ping 109.46.72.6
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.6                                56  64 523us
    1 109.46.72.6                                56  64 546us
    sent=2 received=2 packet-loss=0% min-rtt=523us avg-rtt=534us max-rtt=546us

[admin@R5] > ping 109.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.9                                56  64 31us
    1 109.46.72.9                                56  64 41us
    sent=2 received=2 packet-loss=0% min-rtt=31us avg-rtt=36us max-rtt=41us

[admin@R5] > ping 109.46.72.10
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.10                               56  64 652us
    1 109.46.72.10                               56  64 916us
    sent=2 received=2 packet-loss=0% min-rtt=652us avg-rtt=784us max-rtt=916us
```

```
[admin@R5] > ping 109.46.72.13
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.13                               56  64 163us
    1 109.46.72.13                               56  64 39us
    sent=2 received=2 packet-loss=0% min-rtt=39us avg-rtt=101us max-rtt=163us

[admin@R5] >
```

```
[admin@R5] > ping 109.46.72.13
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.13                               56  64 2ms351us
    1 109.46.72.13                               56  64 40us
    2 109.46.72.13                               56  64 42us
    sent=3 received=3 packet-loss=0% min-rtt=40us avg-rtt=811us max-rtt=2ms351us
```

```
[admin@R6] > ping 109.46.72.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.1                                56  64 1ms961us
    1 109.46.72.1                                56  64 981us
    sent=2 received=2 packet-loss=0% min-rtt=981us avg-rtt=1ms471us max-rtt=1ms961us

[admin@R6] > ping 109.46.72.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.2                                56  64 838us
    1 109.46.72.2                                56  64 739us
    sent=2 received=2 packet-loss=0% min-rtt=739us avg-rtt=788us max-rtt=838us

[admin@R6] > ping 109.46.72.5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.5                                56  64 544us
    1 109.46.72.5                                56  64 605us
    sent=2 received=2 packet-loss=0% min-rtt=544us avg-rtt=574us max-rtt=605us

[admin@R6] > ping 109.46.72.6
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.6                                56  64 65us
    1 109.46.72.6                                56  64 52us
    sent=2 received=2 packet-loss=0% min-rtt=52us avg-rtt=58us max-rtt=65us

[admin@R6] > ping 109.46.72.9
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.9                                56  64 532us
    1 109.46.72.9                                56  64 680us
    sent=2 received=2 packet-loss=0% min-rtt=532us avg-rtt=606us max-rtt=680us

[admin@R6] > ping 109.46.72.10
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 109.46.72.10                               56  64 31us
    1 109.46.72.10                               56  64 44us
    sent=2 received=2 packet-loss=0% min-rtt=31us avg-rtt=37us max-rtt=44us
```

2. Παρατηρούμε ότι υπάρχει η διαδρομή 0.0.0.0/0 σε κάθε interface

```
[admin@R1] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, b, o, d, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS        GATEWAY             DISTANCE
DAd  0.0.0.0/0          192.168.171.2              1
DAc  10.255.255.1/32    loopback0                  0
DAc  108.46.72.0/30     ether2                     0
DAc  108.46.72.4/30     ether3                     0
DAo+ 108.46.72.8/30     108.46.72.6%ether3       110
DAo+ 108.46.72.8/30     108.46.72.2%ether2       110
DAb  109.46.72.8/30     192.168.5.2               20
DAc  192.168.5.0/30     ether4                     0
DAc  192.168.5.1/32     ether4                     0
DAc  192.168.171.0/24   ether1                     0
[admin@R1] >
```

```
[admin@R2] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, s, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
#      DST-ADDRESS       GATEWAY              DISTANCE
  DAo  0.0.0.0/0         108.46.72.1%ether1        110
0   s  0.0.0.0/0         108.46.72.1               112
  DAc  10.255.255.2/32   loopback0                   0
  DAc  108.46.72.0/30    ether1                      0
  DAo+ 108.46.72.4/30    108.46.72.10%ether2       110
  DAo+ 108.46.72.4/30    108.46.72.1%ether1        110
  DAc  108.46.72.8/30    ether2                      0
  DAc  108.46.72.12/30   ether3                      0
  DAo  109.46.72.8/30    108.46.72.1%ether1        110
[admin@R2] >
```

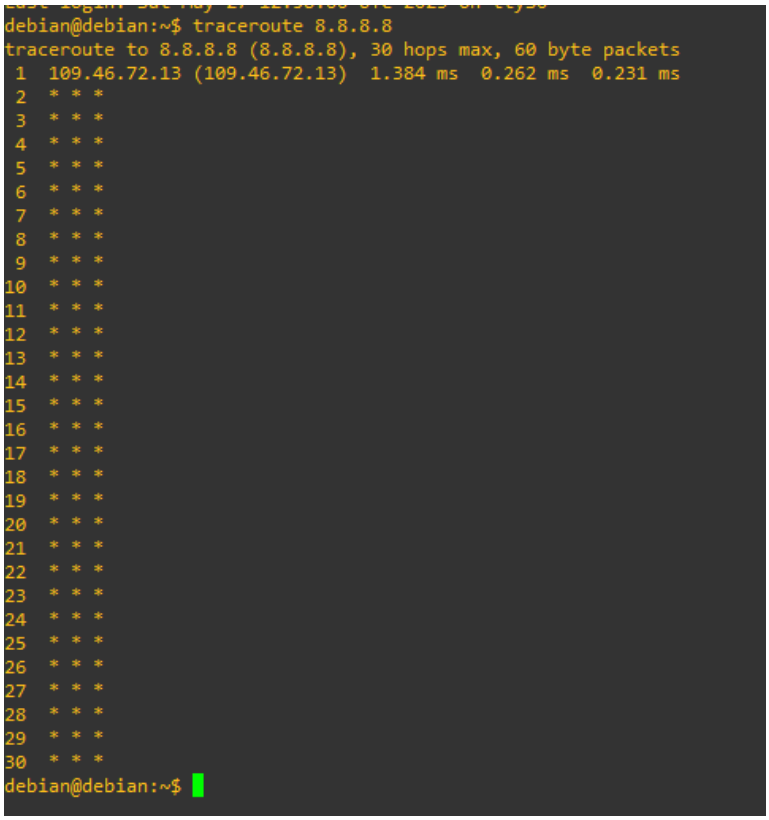
```
[admin@R3] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY              DISTANCE
DAo  0.0.0.0/0         108.46.72.5%ether1        110
DAc  10.255.255.3/32   loopback0                   0
DAo+ 108.46.72.0/30    108.46.72.5%ether1        110
DAo+ 108.46.72.0/30    108.46.72.9%ether2        110
DAc  108.46.72.4/30    ether1                      0
DAc  108.46.72.8/30    ether2                      0
DAo  109.46.72.8/30    108.46.72.5%ether1        110
[admin@R3] >
```

```
[admin@R4] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, b, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY              DISTANCE
DAb  0.0.0.0/0         192.168.5.1                20
DAc  10.255.255.4/32   loopback0                   0
DAb  108.46.72.8/30    192.168.5.1                20
DAc  109.46.72.0/30    ether2                      0
DAc  109.46.72.4/30    ether3                      0
DAo+ 109.46.72.8/30    109.46.72.6%ether3        110
DAo+ 109.46.72.8/30    109.46.72.2%ether2        110
DAc  192.168.5.0/30    ether1                      0
DAc  192.168.5.2/32    ether1                      0
[admin@R4] >
```

```
[admin@R5] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY              DISTANCE
DAo  0.0.0.0/0         109.46.72.1%ether1        110
DAc  10.255.255.5/32   loopback0                   0
DAo  108.46.72.8/30    109.46.72.1%ether1        110
DAc  109.46.72.0/30    ether1                      0
DAo+ 109.46.72.4/30    109.46.72.1%ether1        110
DAo+ 109.46.72.4/30    109.46.72.10%ether2       110
DAc  109.46.72.8/30    ether2                      0
DAc  109.46.72.12/30   ether3                      0
[admin@R5] >
```

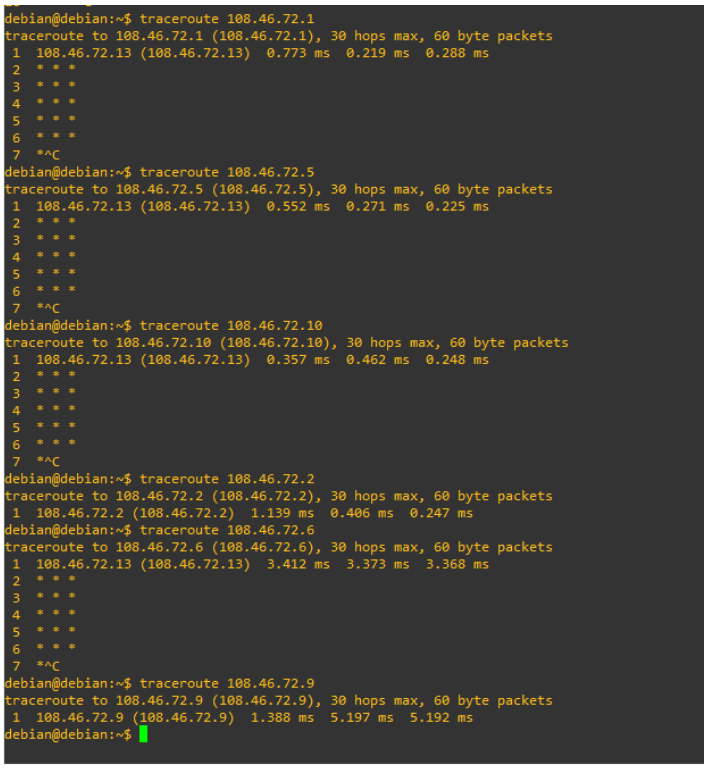
```
[admin@R6] > ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY; + - ECMP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS       GATEWAY              DISTANCE
DAo  0.0.0.0/0         109.46.72.5%ether1        110
DAc  10.255.255.6/32   loopback0                   0
DAo  108.46.72.8/30    109.46.72.5%ether1        110
DAo+ 109.46.72.0/30    109.46.72.9%ether2        110
DAo+ 109.46.72.0/30    109.46.72.5%ether1        110
DAc  109.46.72.4/30    ether1                      0
DAc  109.46.72.8/30    ether2                      0
[admin@R6] >
```

3. Οι διευθύνσεις που βρίσκει το Debian-2 με traceroute 8.8.8.8



4η Ερώτηση

Για το Debian-1



```
debian@debian:~$ traceroute 109.46.72.1
traceroute to 109.46.72.1 (109.46.72.1), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  4.075 ms  3.985 ms  3.977 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.5
traceroute to 109.46.72.5 (109.46.72.5), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  1.614 ms  1.581 ms  1.576 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.2
traceroute to 109.46.72.2 (109.46.72.2), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  0.798 ms  0.236 ms  0.244 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.9
traceroute to 109.46.72.9 (109.46.72.9), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  1.630 ms  1.566 ms  1.559 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.10
traceroute to 109.46.72.10 (109.46.72.10), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  0.973 ms  0.225 ms  0.328 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.6
traceroute to 109.46.72.6 (109.46.72.6), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  0.617 ms  0.304 ms  0.277 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 192.168.5.2
traceroute to 192.168.5.2 (192.168.5.2), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  2.286 ms  2.655 ms  2.649 ms
 2  * * *
 3  * * *
 4  * * *
```

```
debian@debian:~$ traceroute 192.168.5.1
traceroute to 192.168.5.1 (192.168.5.1), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  1.539 ms  1.507 ms  1.502 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$
```

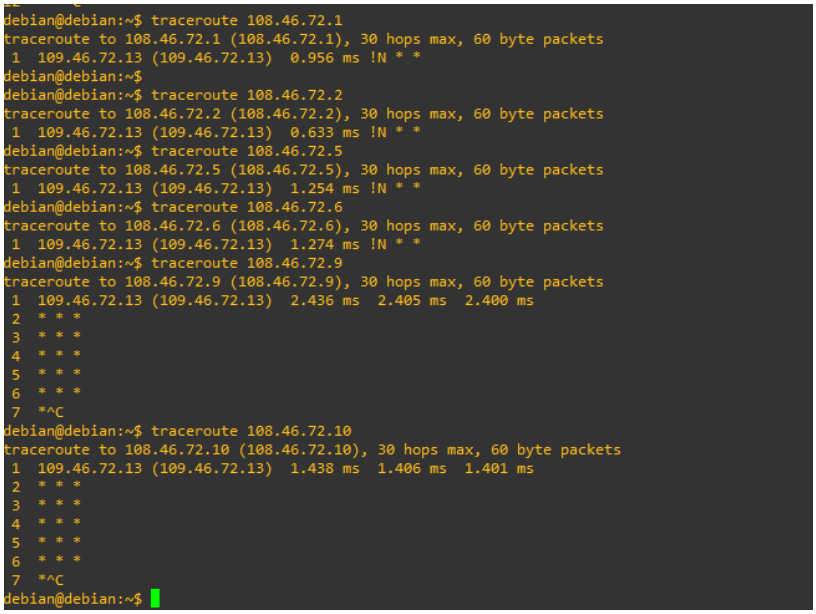
```
debian@debian:~$ traceroute 108.46.72.13
traceroute to 108.46.72.13 (108.46.72.13), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  3.630 ms  3.545 ms  3.532 ms
debian@debian:~$ traceroute 109.46.72.13
traceroute to 109.46.72.13 (109.46.72.13), 30 hops max, 60 byte packets
 1  108.46.72.13 (108.46.72.13)  3.629 ms  5.071 ms  5.612 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  *^C
```

Για το Debian-2

```
debian@debian:~$ traceroute 192.168.5.2
traceroute to 192.168.5.2 (192.168.5.2), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  1.616 ms !N * *
debian@debian:~$ traceroute 192.168.5.1
traceroute to 192.168.5.1 (192.168.5.1), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  1.202 ms !N * *
debian@debian:~$
```

```
debian@debian:~$ traceroute 109.46.72.1
traceroute to 109.46.72.1 (109.46.72.1), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  0.515 ms  0.241 ms  0.314 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.2
traceroute to 109.46.72.2 (109.46.72.2), 30 hops max, 60 byte packets
 1  109.46.72.2 (109.46.72.2)  1.210 ms  0.399 ms  0.236 ms
debian@debian:~$ traceroute 109.46.72.5
traceroute to 109.46.72.5 (109.46.72.5), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  0.851 ms  0.268 ms  0.230 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.6
traceroute to 109.46.72.6 (109.46.72.6), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  0.564 ms  0.239 ms  0.265 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.9
traceroute to 109.46.72.9 (109.46.72.9), 30 hops max, 60 byte packets
 1  109.46.72.9 (109.46.72.9)  3.410 ms  0.290 ms  0.311 ms
debian@debian:~$ traceroute 109.46.72.10
traceroute to 109.46.72.10 (109.46.72.10), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  0.822 ms  0.319 ms  0.294 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$
```

```
debian@debian:~$ traceroute 108.46.72.13
traceroute to 108.46.72.13 (108.46.72.13), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  3.490 ms  15.972 ms  15.866 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *^C
debian@debian:~$ traceroute 109.46.72.13
traceroute to 109.46.72.13 (109.46.72.13), 30 hops max, 60 byte packets
 1  109.46.72.13 (109.46.72.13)  1.544 ms  5.692 ms  6.995 ms
debian@debian:~$
```


debian@debian:~$ traceroute 108.46.72.1
traceroute to 108.46.72.1 (108.46.72.1), 30 hops max, 60 byte packets
1 109.46.72.13 (109.46.72.13) 0.956 ms !N * *
debian@debian:~$
debian@debian:~$ traceroute 108.46.72.2
traceroute to 108.46.72.2 (108.46.72.2), 30 hops max, 60 byte packets
1 109.46.72.13 (109.46.72.13) 0.633 ms !N * *
debian@debian:~$ traceroute 108.46.72.5
traceroute to 108.46.72.5 (108.46.72.5), 30 hops max, 60 byte packets
1 109.46.72.13 (109.46.72.13) 1.254 ms !N * *
debian@debian:~$ traceroute 108.46.72.6
traceroute to 108.46.72.6 (108.46.72.6), 30 hops max, 60 byte packets
1 109.46.72.13 (109.46.72.13) 1.274 ms !N * *
debian@debian:~$ traceroute 108.46.72.9
traceroute to 108.46.72.9 (108.46.72.9), 30 hops max, 60 byte packets
1 109.46.72.13 (109.46.72.13) 2.436 ms 2.405 ms 2.400 ms
2 * * *
3 * * *
4 * * *
5 * * *
6 * * *
7 *^C
debian@debian:~$ traceroute 108.46.72.10
traceroute to 108.46.72.10 (108.46.72.10), 30 hops max, 60 byte packets
1 109.46.72.13 (109.46.72.13) 1.438 ms 1.406 ms 1.401 ms
2 * * *
3 * * *
4 * * *
5 * * *
6 * * *
7 *^C
debian@debian:~$